

Checkpoint 6: Parameter-Matched Capacity Control

Objective

Check whether the main architecture comparison is explained by parameter count.

Key Message

The capacity control makes the final claim honest: matched U-Net leads pixel metrics while residual leads mean SSIM and Delta E.

Completed Work

- Completed parameter-matched U-Net capacity control with base_channels=42.
- Compared parameter-matched U-Net with the residual model.
- Qualified the final conclusion using metric-specific capacity-control evidence.

Verification and Controls

- Matched trainable capacity while retaining the plain U-Net topology to isolate architecture from model size.
- Confirmed metric direction explicitly: lower for MSE, MAE, and Delta E; higher for PSNR and SSIM.
- Recorded mixed leadership and seed variability instead of selecting a universal winner.

Evidence

Metric	Matched U-Net	Residual	Mean leader
MSE	0.038056 +/- 0.002727	0.038782 +/- 0.003119	Matched U-Net
PSNR	16.017719 +/- 0.277610	16.006089 +/- 0.267086	Matched U-Net
SSIM	0.722221 +/- 0.022096	0.779948 +/- 0.006607	Residual
Delta E	27.590478 +/- 2.280969	27.053617 +/- 1.081249	Residual
Entropy	4.946796 +/- 0.060048	4.565555 +/- 0.202586	Descriptive only

Decision / Next Gate

Conclude that matched U-Net leads mean pixel metrics while the residual model leads mean SSIM and Delta E.

Notes

- The residual model leads on SSIM and Delta E; the matched U-Net leads on MSE, MAE, and PSNR.



Matching parameter-matched U-Net and residual restoration examples.